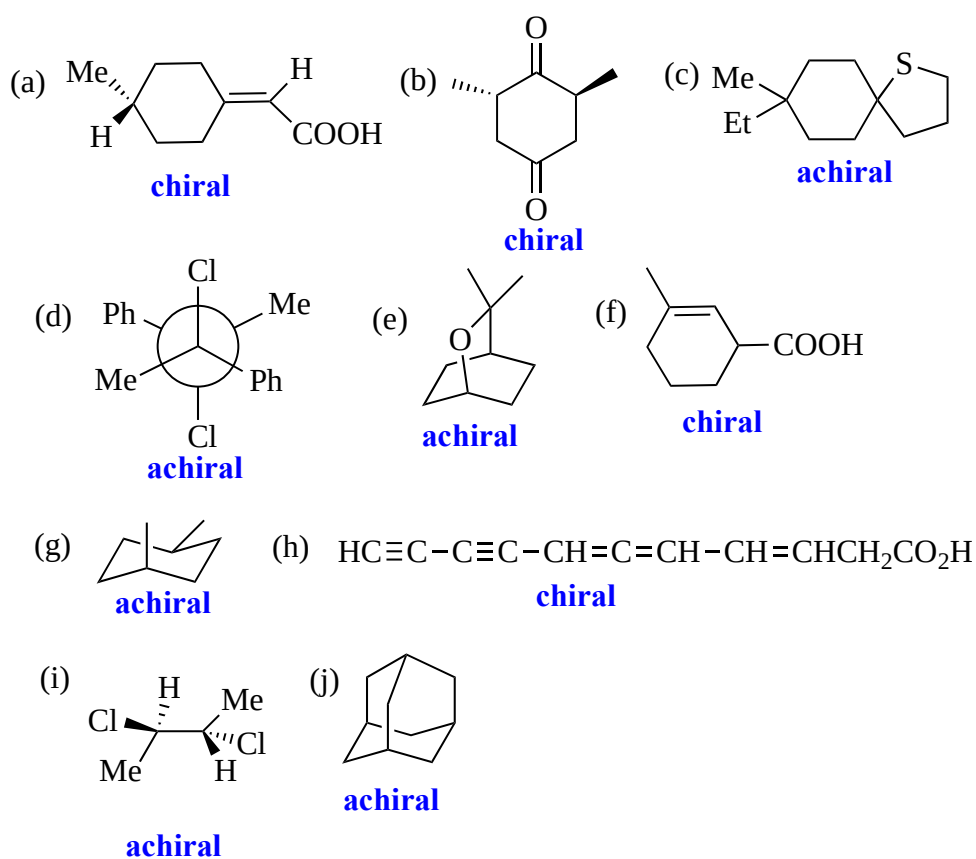


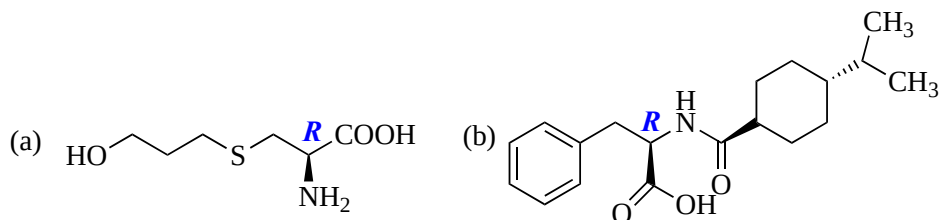
Department of Chemistry
CML 101: Organic Chemistry: Tutorial 1

Instructor: Dr. Nidhi Jain (Room: MS 727)

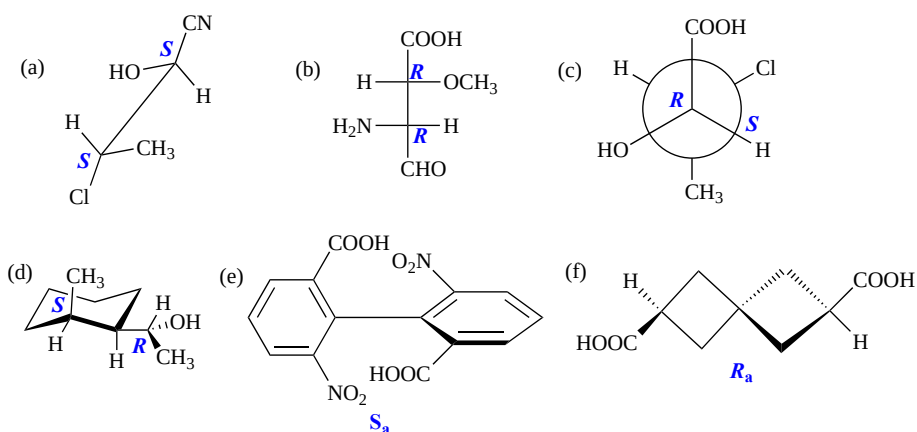
1. Identify the following molecules as chiral or achiral?



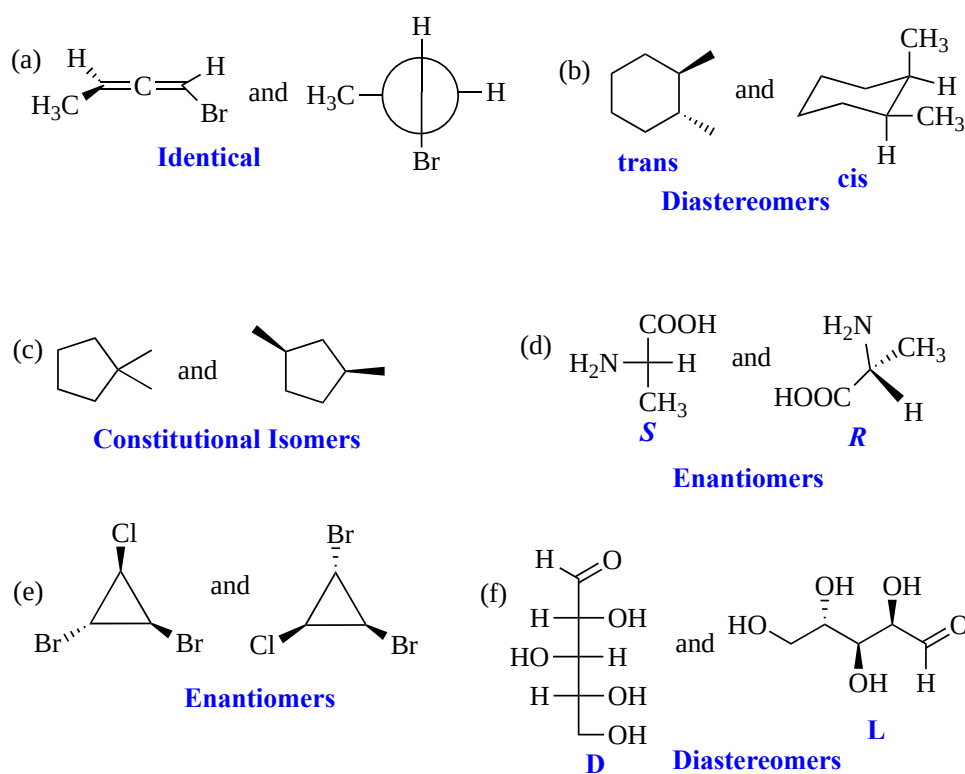
2. Assign R/S configuration to the following compounds



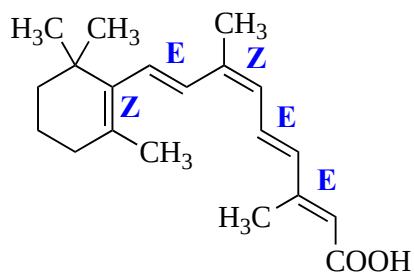
3. Determine the absolute configuration of the following compounds



4. What is the relationship between the following pair of compounds?

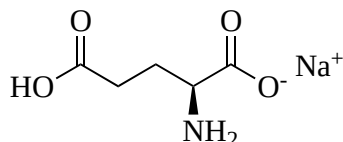


5. Assign E/Z configuration to the double bonds in the compound given below. How many stereo isomers are possible?



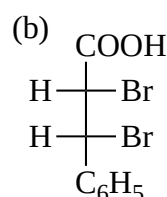
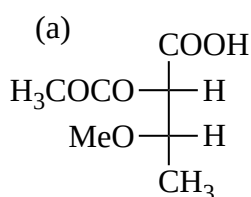
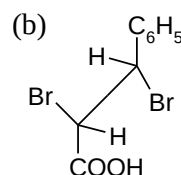
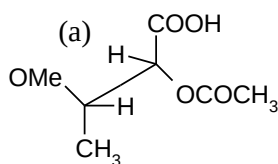
Theoretical number of isomers (x) = 2^n where n = number of stereogenic units hence $2^4 = 16$. Although the molecule contains five double bonds, only those in the side chain can be considered, since because of ring strain the double bond in the six membered ring can only have Z-configuration.

6. Pure MSG, a common flavor enhancer, exhibits a specific rotation of +24. (a) Calculate the *ee* of a solution whose $[\alpha]$ is +10. (b) If the *ee* of a solution is 80%, what is the $[\alpha]$ for this solution?



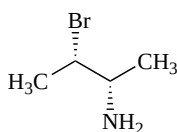
(a) 42% *ee* (b) $[\alpha] = +19.2$

7. Draw the following molecules in their respective Fischer projection formula

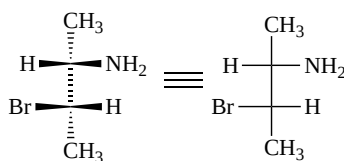


Fischer projection

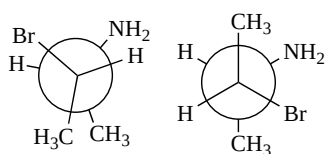
8. Draw the Wedge, Fischer, Newman and Sawhorse projections for the **(2S,3S)-3-bromobutan-2-amine**.



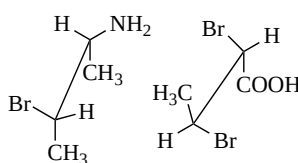
(2S,3S)-3-bromobutan-2-amine.



Fischer Projection



Newman Projection



Sawhorse Projection